Biotech Techniques

The applications/uses of biotechnology tools

Biotechnology – vocabulary



- Amplifying
- Annealing
- DNA ladder
- Genome
- Plasmid
- Micropipette
- Short Tandem Repeat (STR)
- Vector

Table 6.4 on p175-177 Card sort activity

Biotechnology tools



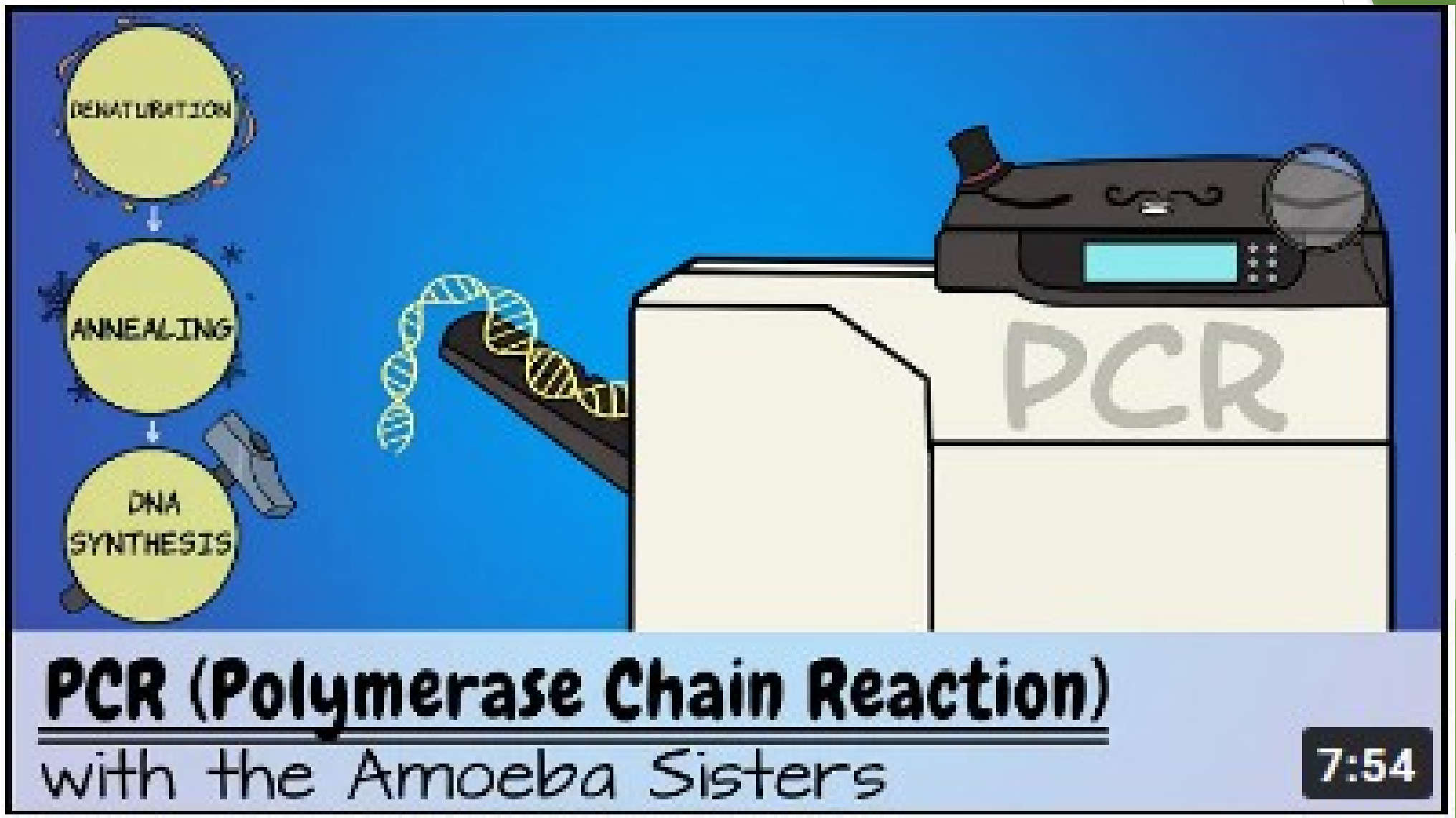
- PCR
- Gel electrophoresis
- DNA sequencing
- Recombinant DNA
- Microarrays

PCR – Amplifying DNA



- Polymerase Chain Reaction
- Technique used to rapidly replicate small amounts of DNA
  - E.g. From a crime scene, from bones
- Amplified DNA used later in other techniques such as gel electrophoresis or DNA profiling

Amoeba Sisters Video



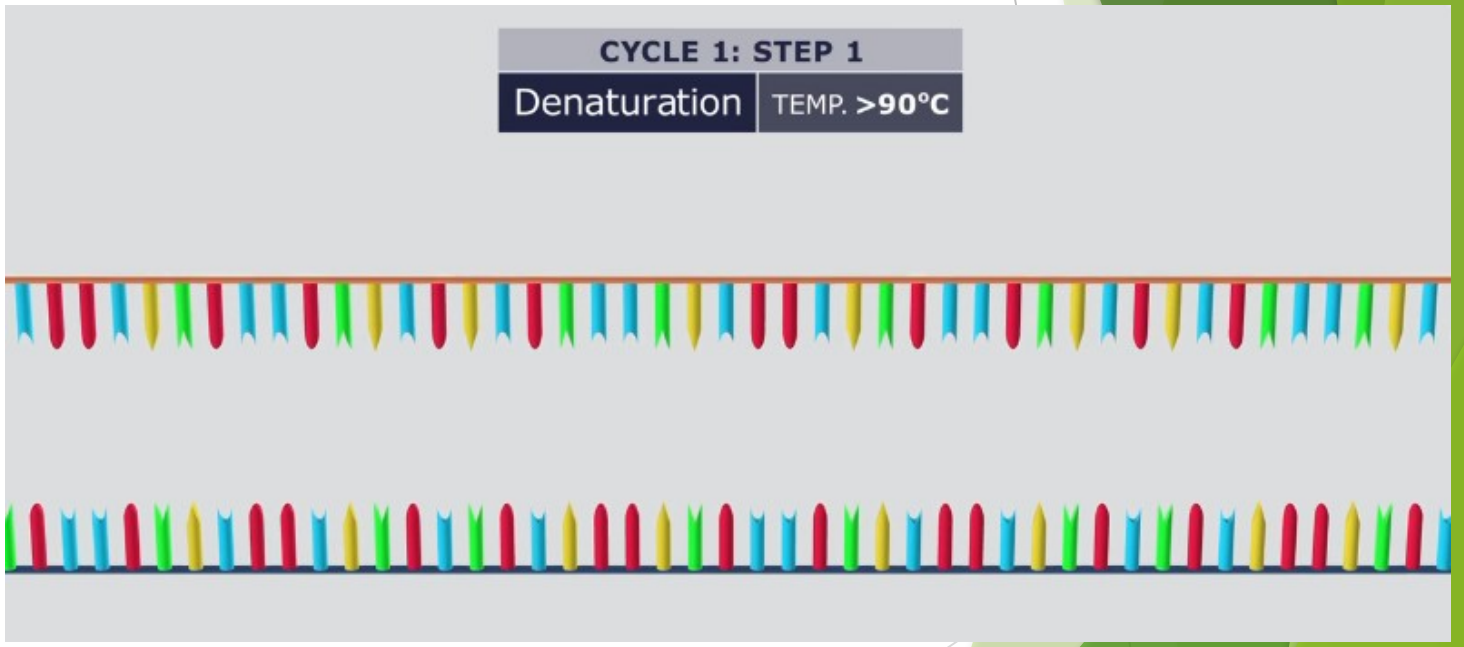
PCR – Amplifying DNA



- Ingredients:
  - Original DNA
  - Taq polymerase
  - Buffer solution
  - Free nucleotides
  - DNA primers

Step 1 – Denaturing DNA

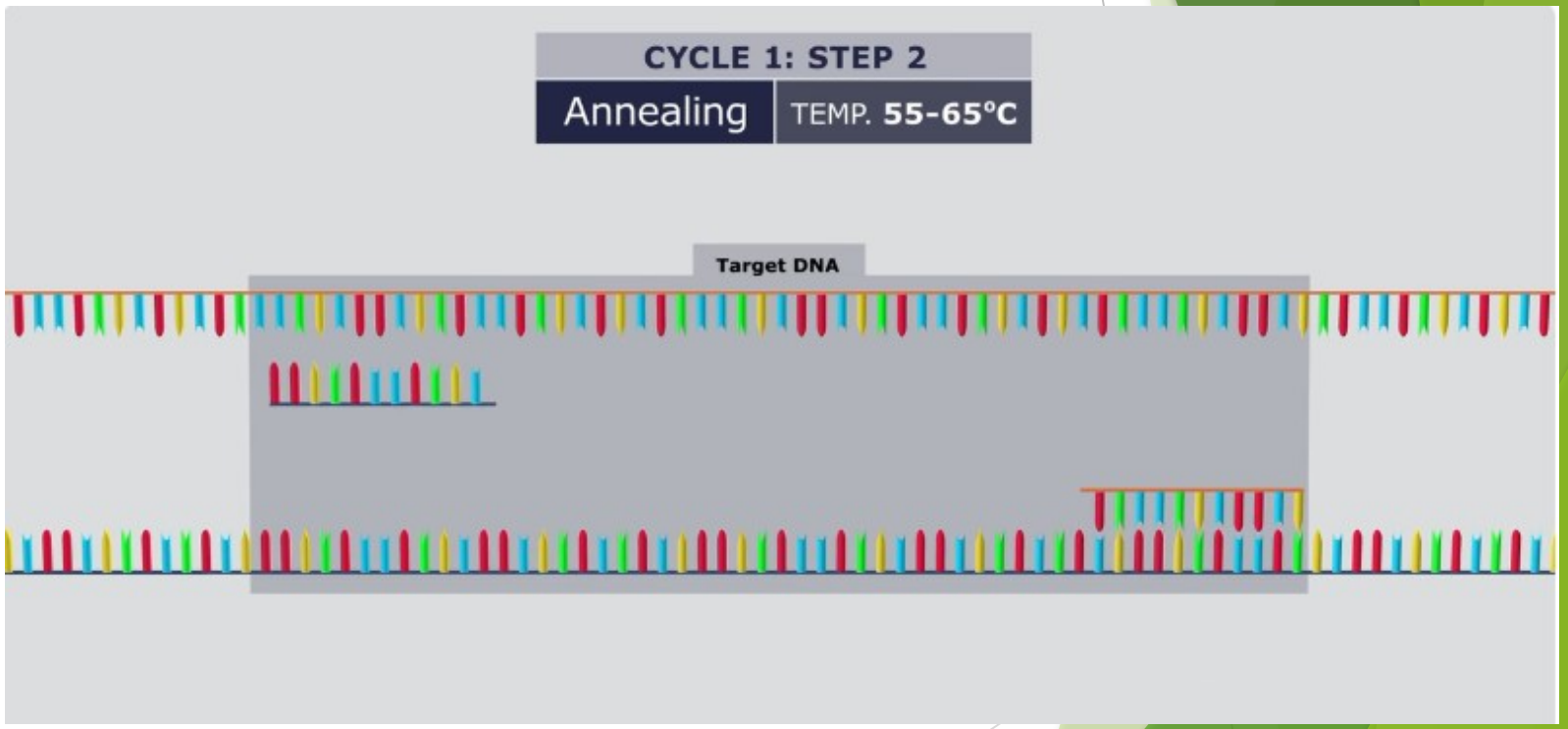
- Heat sample DNA to 95°C
- Breaks H-bonds between bases
- Separates strands



Step 2 – Annealing

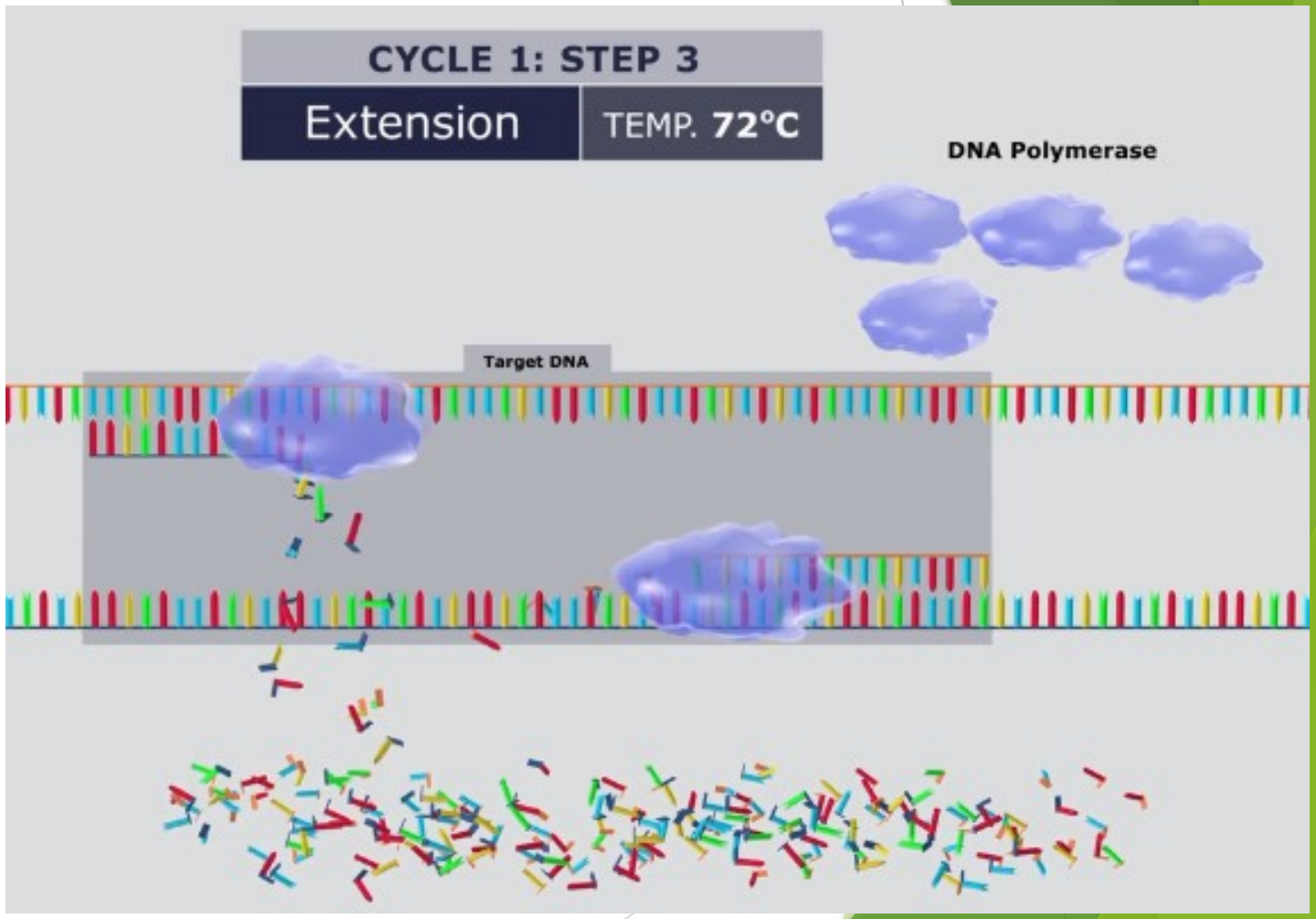


- Temperature lowered to 55-65°C
- Primers bind to ends of target DNA sequence



Step 3 – Extension

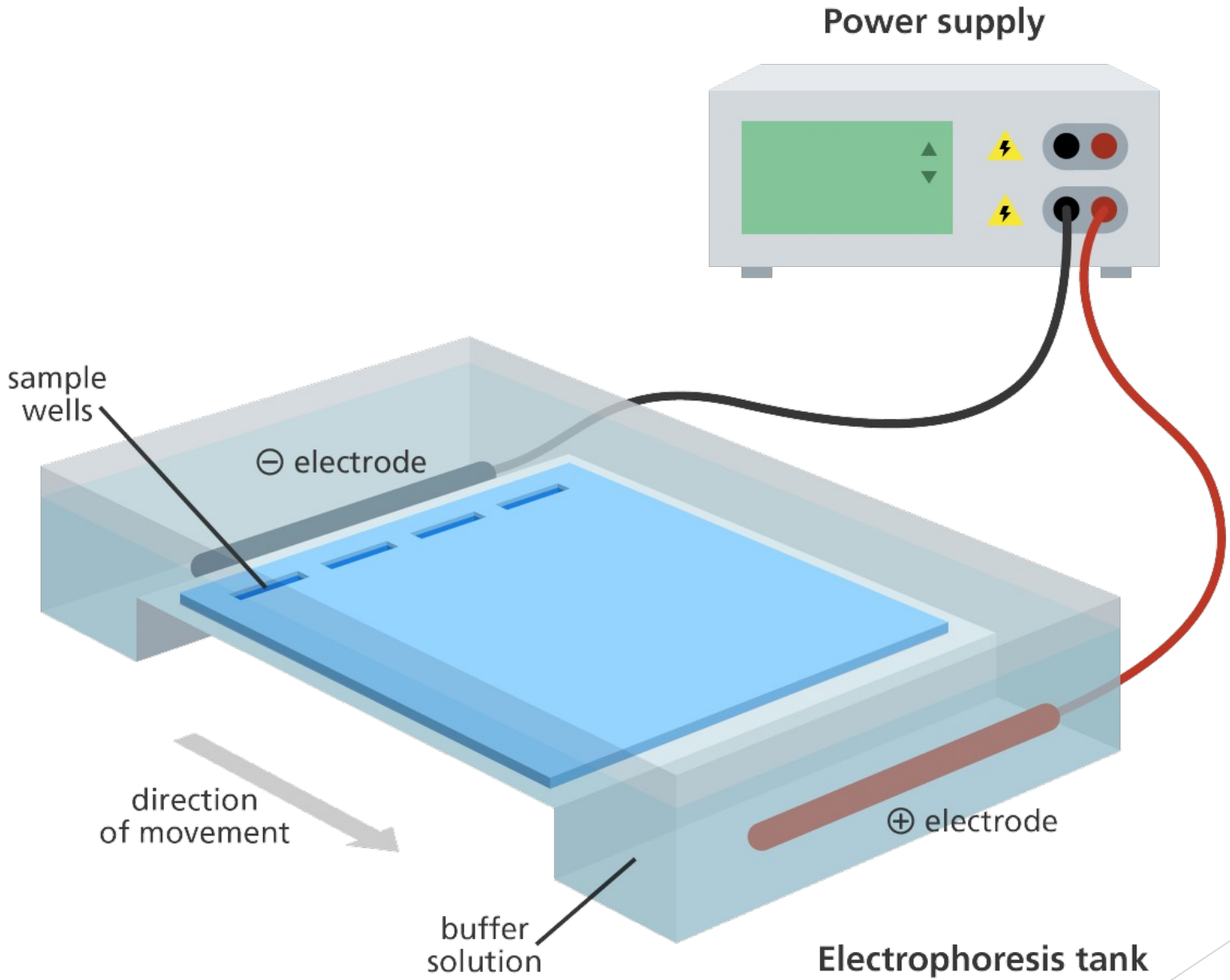
- Temperature 72ºC, optimal for Taq polymerase enzyme
- Attaches to primers, moves along strand, attaching free nucleotides
- Ends with twice amount of DNA



Activities



- Nelson textbook QS 6.3a, p180

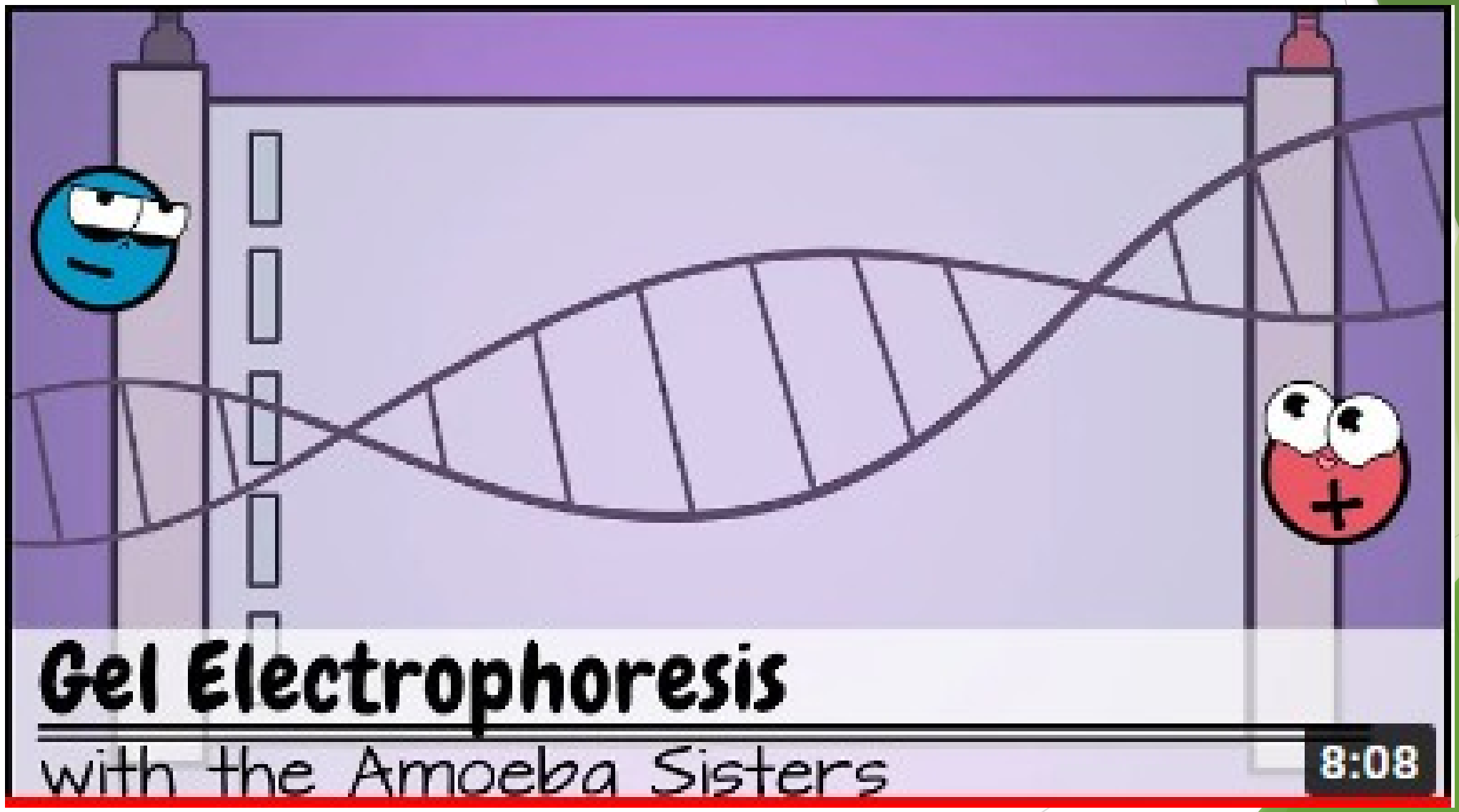

Power supply
sample wells
⊖ electrode
direction of movement
⊕ electrode
buffer solution
Electrophoresis tank

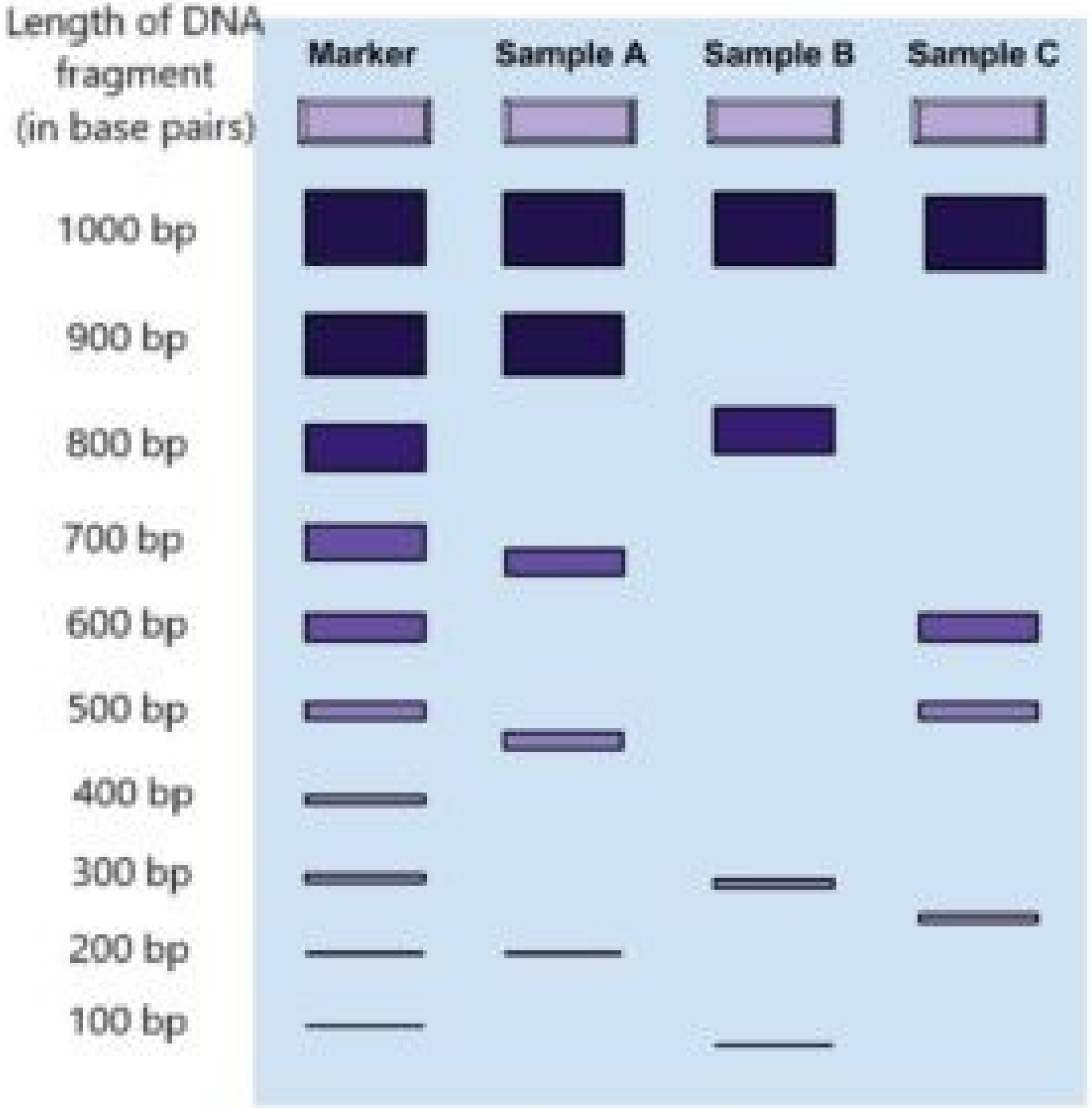
Gel electrophoresis



- Way of visualising DNA and comparing samples
- Samples of DNA are cut into fragments using REs
- Loaded into wells on agarose gel
- Negatively charged fragments drawn through gel by electric current
- Smaller fragments move faster and further

Amoeba Sister Video




Length of DNA fragment (in base pairs)
Marker
Sample A
Sample B
Sample C
1000 bp
900 bp
800 bp
700 bp
600 bp
500 bp
400 bp
300 bp
200 bp
100 bp

Activities

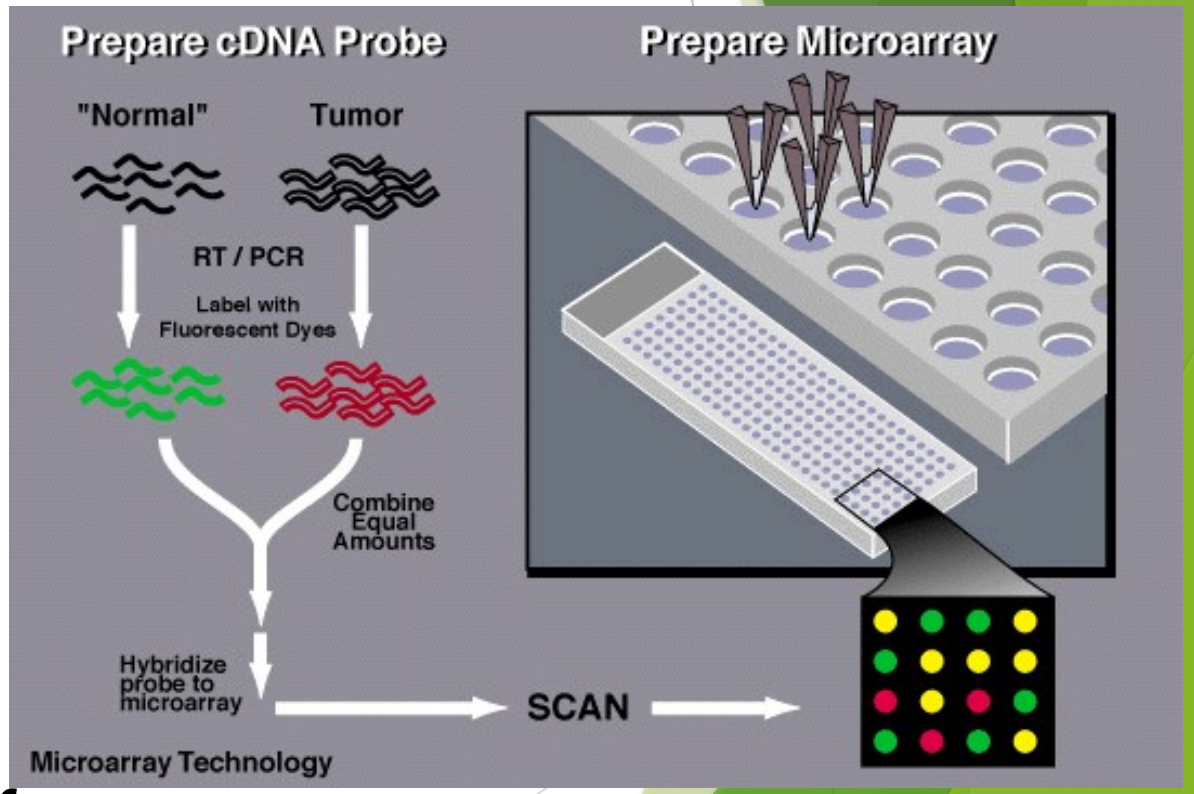




- Nelson textbook QS 6.3b, p183-184
- Nelson textbook Activity 6.1, p199-200 and 212, “Brood parasitism and family size in black swans”

Microarrays – Probing for genes

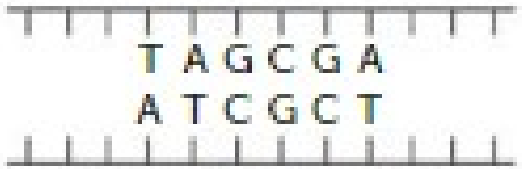
- Tool used to determine if DNA from an individual contains a particular gene expression
- Probe is a short sequence of DNA made radioactive
- Target sequence is complementary to probe
- Probe attaches and fluoresces to show presence of gene
- E.g. BRCA1 and BRCA2 genes for breast cancer



1 A probe is a sequence of DNA that is made radioactive.



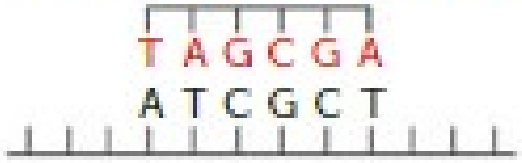
2 The target for the probe is double-stranded DNA containing the sequence being studied.



3 The target DNA is heat-treated to separate the strands.

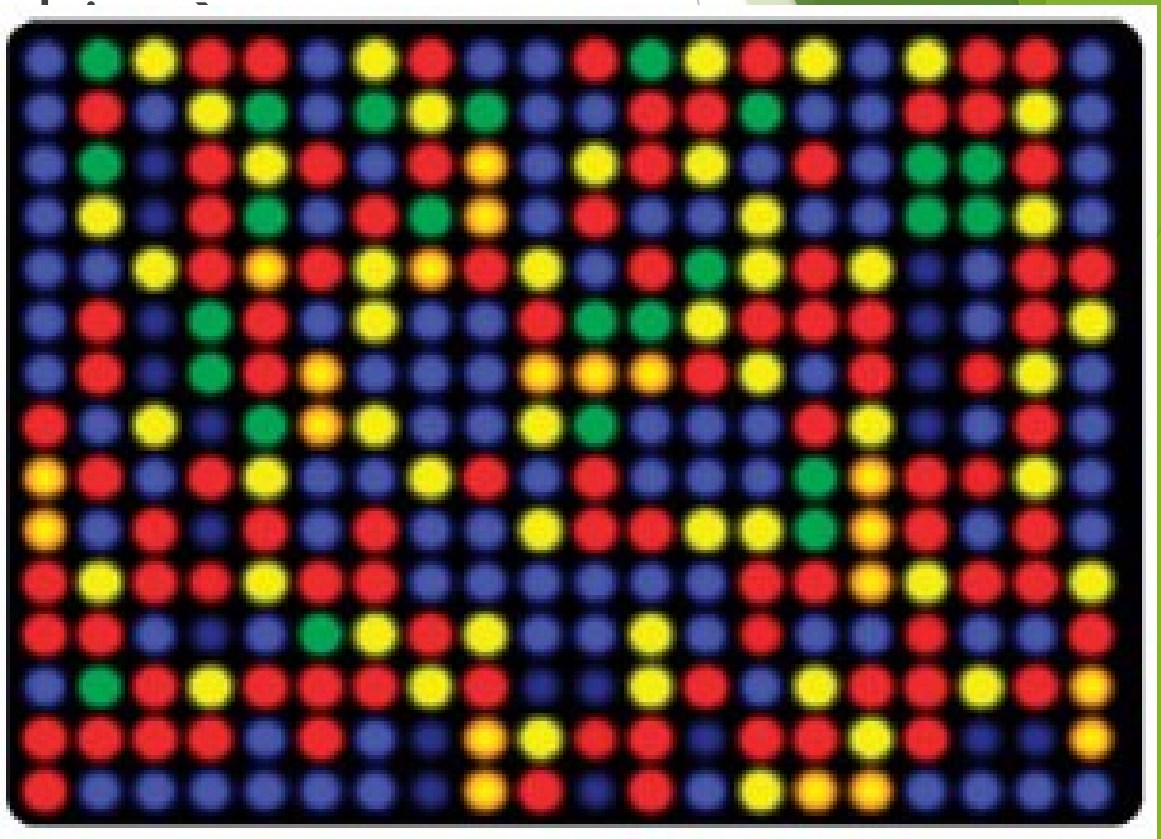


4 The radioactive probe is introduced to find the gene.



Microarrays – probing for genes

- Thousands of DNA probes arrayed on a single glass microscope slide or silicon
- Probes allowed to bind to genes
- A scanner measures fluorescence
- Analyse to find gene activity
  - Brighter colour, more activity



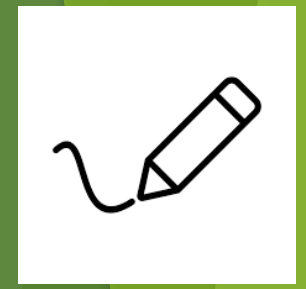
More info at BioNinja

Activities



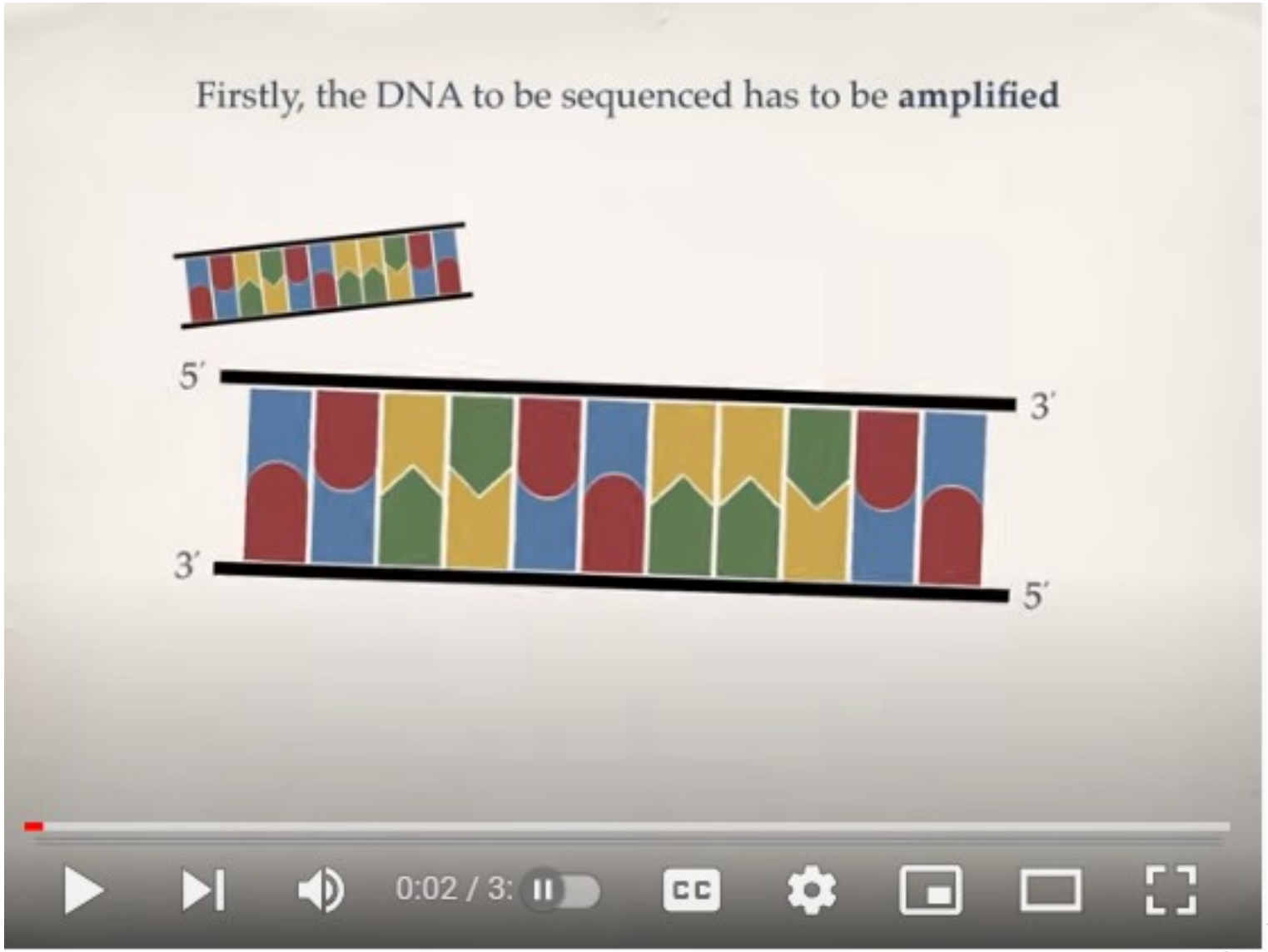
- Nelson textbook QS 6.3c, p185

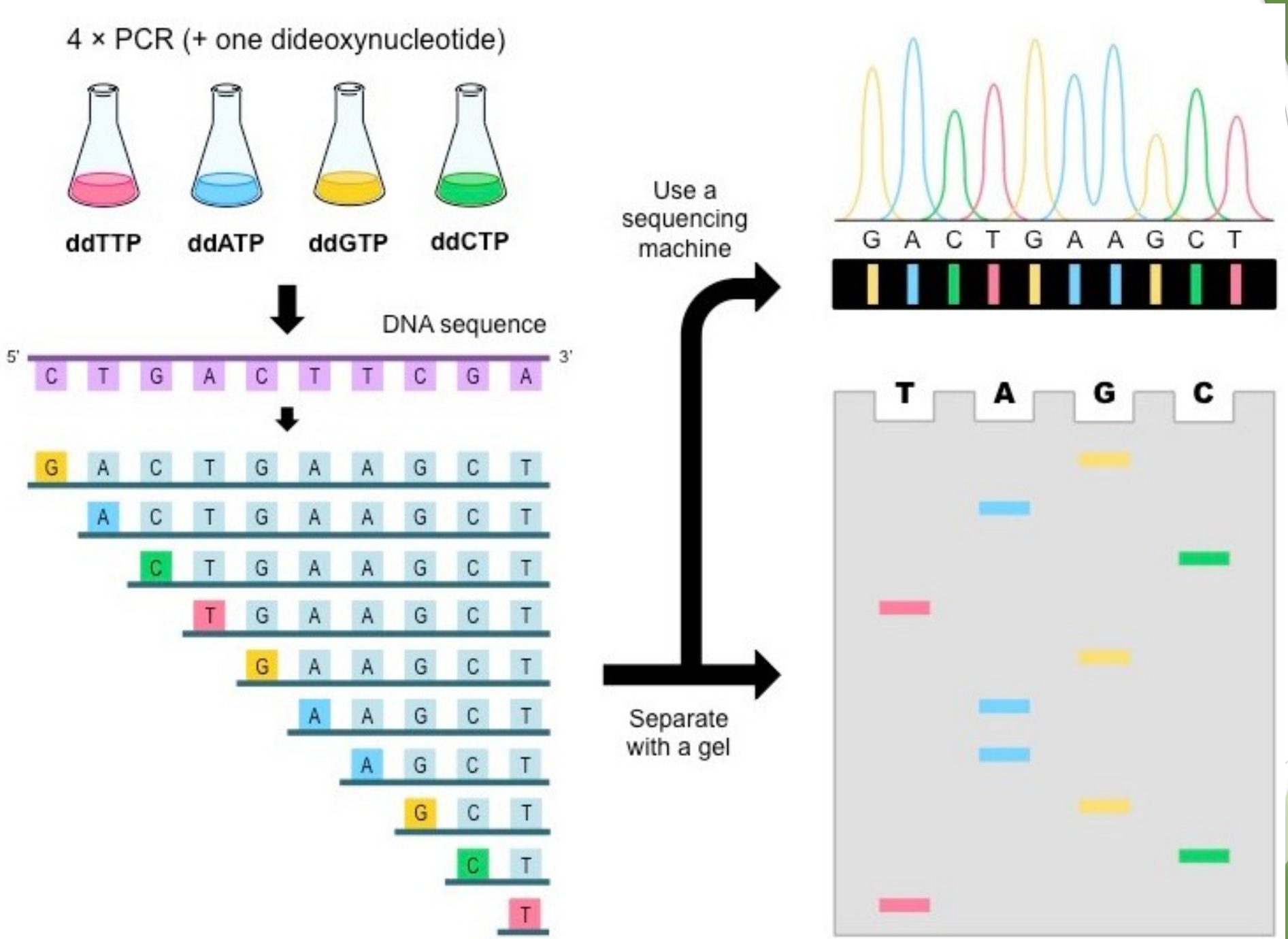
DNA sequencing



- Refers to the methods and technologies used to determine the exact order of nucleotide bases in DNA fragments
- Useful to identify abnormal gene function, future applications in conservation and agriculture
- Two methods:
  1. Sanger method
  2. Next-generation sequencing

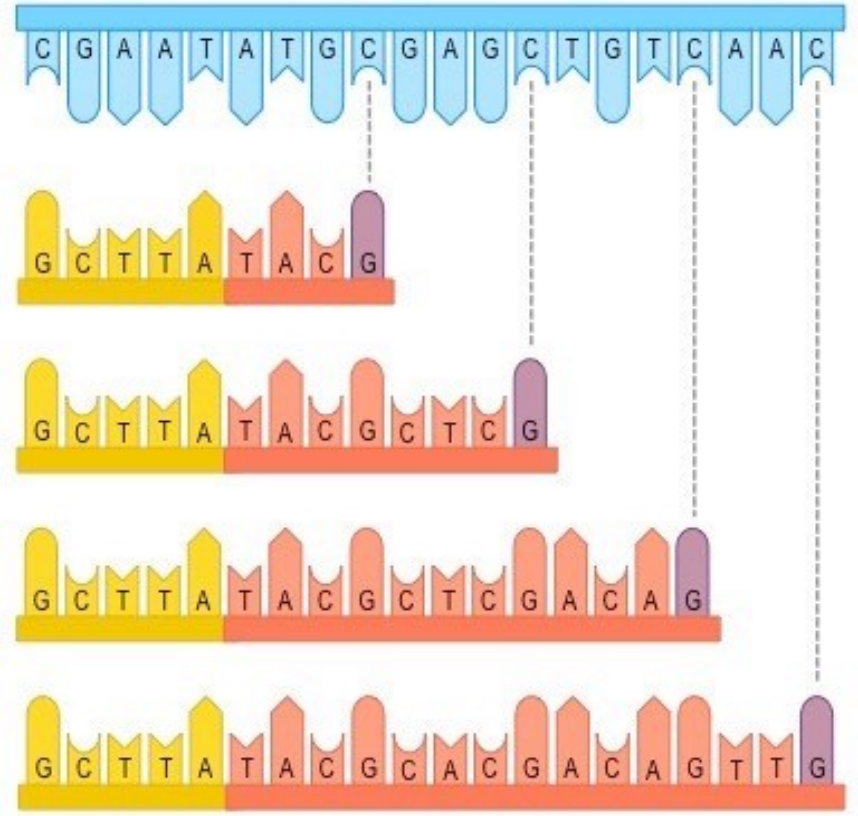
DNA sequencing – Sanger method



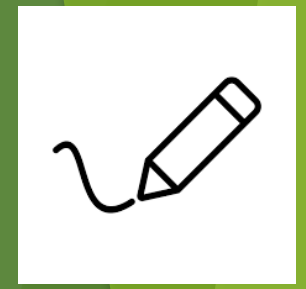

4 × PCR (+ one dideoxynucleotide)
ddTTP
ddATP
ddGTP
ddCTP
DNA sequence
5'
3'
C T G A C T T C G A
G A C T G A A G C T
A C T G A A G C T
C T G A A G C T
T G A A G C T
G A A G C T
A A G C T
A G C T
G C T
C T
T
Use a sequencing machine
Separate with a gel
G A C T G A A G C T
T A G C

DNA sequencing – Sanger method

- Four PCR mixtures are made up, containing normal nucleotides PLUS one dideoxynucleotide (ddA, ddT, ddC, ddG)
- ddNTPs are modified nucleotides that integrate into the sequence but prevent the addition of anything after it →
  - Terminates sequence causing fragments



DNA sequencing – Sanger method



- PCR generates billions of copies of DNA, so will make all possible fragment lengths
- Fragments are separated using gel electrophoresis
- If a dye is added, fragments can be read by a machine

DNA sequencing – NGS



- More advanced than Sanger method

1. DNA prepared: Broken into fragments ~300bp then amplified by PCR
2. Sequencing: On gel plate, each well contains ddNTPs with fluorescent tags added to fragments
3. Data analysis: Flashes of light reveal the sequence of nucleotides – a different colour for each base – and software identifies the order

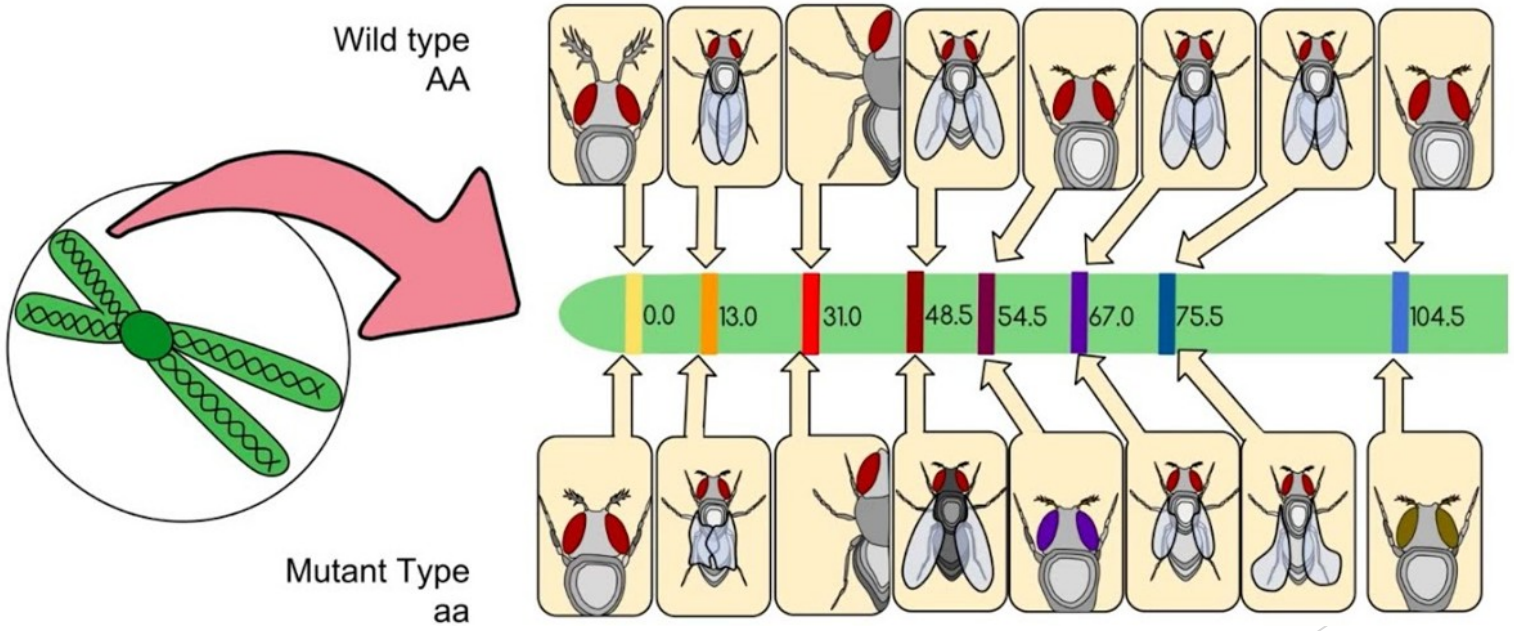
Activities



- Application 6.1 “Comparative genomics between humans and fruit flies”, p188
- Case Study “Use of next-generation sequencing to study ecosystem biodiversity”, p188-189
- QS 6.4a, p190

DNA sequencing – Genome mapping

- Genome – entire set of genetic instructions (both coding genes and non-coding)
- Genetic mapping identifies and records the position of genes on a chromosome (locus)



DNA sequencing – Genome mapping video



Activities



- QS 6.4b, p191
- Scientific Literacy “Using genetic markers in the quest to save our vulnerable burrowing bettong”, p192-193

DNA profiling



- Also known as DNA fingerprinting
- The process used to compare the base sequence of two or more individuals to determine how closely related they are
- Most DNA is identical except for specific regions called polymorphic regions
- Unique combination of polymorphic regions inherited from parents

DNA profiling



- Short Tandem Repeats (STRs) – regions of non-coding DNA that repeats same sequence
  - E.g. GATAGATAGATAGATAGATAGATAGATAGATAGATA
- Each individual has unique number of repeats for each gene
- 10+ STR loci compared to produce a DNA profile
- Use PCR and gel electrophoresis to make profiles and compare

Activities



- QS 6.5, p194
- Application 6.2, “Superb fairy wrens – not as faithful as you thought!”, p194